# Dual Simplex algorithm

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# Pair of primal and dual problems

# Introduction: Pair of primal and dual problems

- The dual simplex algorithm works on the same dictionaries as the simplex algorithm, but to understand it we look at the **primal** and the **dual** problem together
- We write the dual problem in standard maximization form, adding the surplus variables *before* the dual variables
- In this way every primal variable has a dual variable with the same index: this correspondence is used in the whole chapter
- We also introduce the running example of the chapter, an LP problem with $n = 2$ variables and $m = 3$ constraints

# The dual problem in standard maximization form

- We consider the standard maximization form of the dual problem, obtained by adding $n$ surplus non-negative variables to a dual LP problem with $m$ non-negative variables and $n$ greater-than-or-equal-to linear constraints:

$$\begin{aligned} \max \quad & -\boldsymbol{b}'\,\boldsymbol{\pi} \\ \text{subject to} \quad & \boldsymbol{A}'\,\boldsymbol{\pi} = \boldsymbol{c}, \\ & \boldsymbol{\pi} \geq 0. \end{aligned}$$

# Surplus variables first

- The $n$ non-negative surplus variables are added *before* the $m$ dual variables in order to shift the indices of the dual variables by $n$
- We obtain the following index correspondence:
  - the $n$ surplus variables $\pi_1, \pi_2, \ldots, \pi_n$ of the dual problem can be associated to the $n$ primal variables $x_1, \ldots, x_n$;
  - the $m$ dual variables $\pi_{n+1}, \pi_{n+2}, \ldots, \pi_{n+m}$ of the dual problem can be associated to the $m$ slack variables $x_{n+1}, \ldots, x_{n+m}$ of the primal problem.

# Input data of the dual problem (1/3)

The input data of the dual problem are:

$$\boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \\ b_{n+1} \\ b_{n+2} \\ \vdots \\ b_{n+m} \end{pmatrix} \qquad \boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

# Input data of the dual problem (2/3)

. . . and the matrix

$$\boldsymbol{A}' = \begin{pmatrix} a'_{11} & a'_{12} & \cdots & a'_{1n} & a'_{1(n+1)} & a'_{1(n+2)} & \cdots & a'_{1(n+m)} \\ a'_{21} & a'_{22} & \cdots & a'_{2n} & a'_{2(n+1)} & a'_{2(n+2)} & \cdots & a'_{2(n+m)} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a'_{n1} & a'_{n2} & \cdots & a'_{nn} & a'_{n(n+1)} & a'_{n(n+2)} & \cdots & a'_{n(n+m)} \end{pmatrix}$$

- $n$ rows, one for each constraint of the dual problem
- $n + m$ columns, one for each variable of the dual problem

# Input data of the dual problem (3/3)

The $n + m$ non-negative dual variables can be written as a column vector $\boldsymbol{\pi}$ of $n + m$ rows:

$$\boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_n \\ \pi_{n+1} \\ \pi_{n+2} \\ \vdots \\ \pi_{n+m} \end{pmatrix}$$

# The columns of the surplus variables

Since the dual problem is obtained by adding $n$ surplus variables to a dual LP problem with $n$ greater-than-or-equal-to constraints, we have:

$$\begin{pmatrix} a'_{11} & a'_{12} & \cdots & a'_{1n} \\ a'_{21} & a'_{22} & \cdots & a'_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a'_{n1} & a'_{n2} & \cdots & a'_{nn} \end{pmatrix} = \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}$$

which is the negative identity matrix $-\boldsymbol{I}$ of size $n$. Moreover, for the same reason, we have:

$$\begin{pmatrix} b_1 & b_2 & \cdots & b_n \end{pmatrix}' = \begin{pmatrix} 0 & 0 & \cdots & 0 \end{pmatrix}'$$

Let us now see the pair of problems
on the running example of the chapter:

an LP problem in canonical maximization form
with $n = 2$ variables and $m = 3$ constraints.

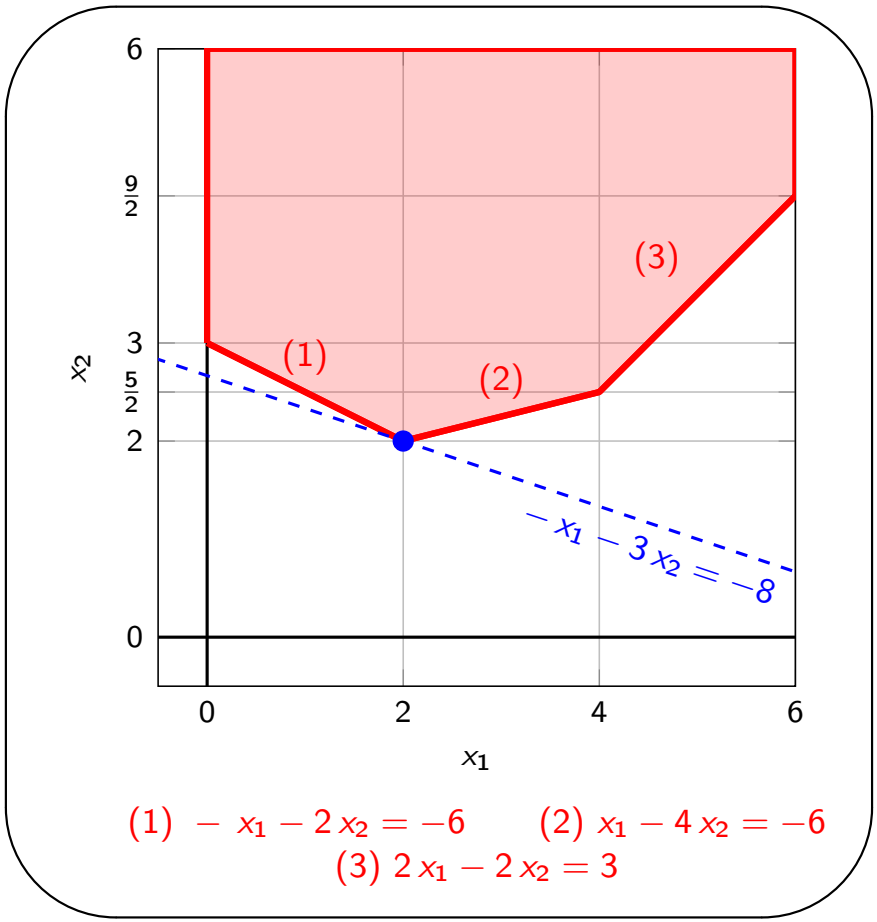
# Example (1/6)

- We consider the following LP problem in canonical maximization form with $n = 2$ variables and $m = 3$ constraints:

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & \\
\text{subject to} & - & x_1 & - & 2 & x_2 & \leq & -6 \\
 & & x_1 & - & 4 & x_2 & \leq & -6 \\
 & 2 & x_1 & - & 2 & x_2 & \leq & 3 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

- We call this LP problem the **running example**: we will use it in the whole chapter

# Example (2/6)




An optimal solution $\tilde{\boldsymbol{x}}(\, LP\,)$ is:

$$\tilde{x}_1 = 2, \quad \tilde{x}_2 = 2$$

and the optimal value is:

$$z(\, LP\,) = -8$$

# Example (3/6)

- Adding $m = 3$ slack variables, the standard form of the running example with $n + m = 5$ variables and $m = 3$ constraints reads as follows:

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & & & & & & & & \\
\text{subject to} & - & x_1 & - & 2 & x_2 & + & x_3 & & & & & = & -6 \\
 & & x_1 & - & 4 & x_2 & & & + & x_4 & & & = & -6 \\
 & 2 & x_1 & - & 2 & x_2 & & & & & + & x_5 & = & 3 \\
 & & x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

# Example (4/6)

- The dual problem of the running example is the following LP problem in canonical minimization form with $m = 3$ variables and $n = 2$ constraints:

$$
\begin{array}{rrrrrrrrrrr}
\text{min} & -6 & \pi_1 & - & 6 & \pi_2 & + & 3 & \pi_3 & & \\
\text{subject to} & - & \pi_1 & + & & \pi_2 & + & 2 & \pi_3 & \geq & -1 \\
& -2 & \pi_1 & - & 4 & \pi_2 & - & 2 & \pi_3 & \geq & -3 \\
& & \pi_1, & & & \pi_2, & & & \pi_3 & \geq & 0
\end{array}
$$

# Example (5/6)

- Shifting the indices of the dual variables by $n = 2$ and adding $n = 2$ surplus variables, we then obtain the following standard maximization form:

$$
\begin{array}{lrrrrrrrrr}
\text{max} & & & & & 6\,\pi_3 & + & 6\,\pi_4 & - & 3\,\pi_5 \\
\text{subject to} & -\,\pi_1 & & - & & \pi_3 & + & \pi_4 & + & 2\,\pi_5 = -1 \\
& & -\,\pi_2 & - & 2\,\pi_3 & - & 4\,\pi_4 & - & 2\,\pi_5 = -3 \\
& \pi_1, & \pi_2, & & \pi_3, & & \pi_4, & & \pi_5 \geq & 0
\end{array}
$$

- The surplus variables $\pi_1, \pi_2$ correspond to $x_1, x_2$, and the dual variables $\pi_3, \pi_4, \pi_5$ correspond to the slack variables $x_3, x_4, x_5$

# Example (6/6)

- In this dual problem in standard form, we have:

$$\boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ -6 \\ -6 \\ 3 \end{pmatrix} \quad \boldsymbol{A}' = \begin{pmatrix} -1 & 0 & -1 & 1 & 2 \\ 0 & -1 & -2 & -4 & -2 \end{pmatrix} \quad \boldsymbol{c} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$$

- and the vector of the variables is

$$\boldsymbol{\pi} = \begin{pmatrix} \pi_1 & \pi_2 & \pi_3 & \pi_4 & \pi_5 \end{pmatrix}'$$

- An optimal solution $\tilde{\boldsymbol{\pi}}(\, D(LP)\,)$ of this LP problem is:

$$\tilde{\pi}_1 = 0, \quad \tilde{\pi}_2 = 0, \quad \tilde{\pi}_3 = \frac{7}{6}, \quad \tilde{\pi}_4 = \frac{1}{6}, \quad \tilde{\pi}_5 = 0$$

# Summary: Pair of primal and dual problems

**The dual problem in standard maximization form**:

- max $-\boldsymbol{b}'\boldsymbol{\pi}$ subject to $\boldsymbol{A}'\boldsymbol{\pi} = \boldsymbol{c}$, $\boldsymbol{\pi} \geq 0$
- The $n$ surplus variables come first: their columns form $-\boldsymbol{I}$ and their coefficients in $\boldsymbol{b}$ are zero

**Index correspondence**:

- $\pi_1, \ldots, \pi_n \leftrightarrow x_1, \ldots, x_n$ (primal variables)
- $\pi_{n+1}, \ldots, \pi_{n+m} \leftrightarrow x_{n+1}, \ldots, x_{n+m}$ (slack variables)

**Running example**:

- Optimal solution $\tilde{x}_1 = \tilde{x}_2 = 2$, optimal value $-8$; dual optimal solution $\tilde{\pi}_3 = \frac{7}{6}$, $\tilde{\pi}_4 = \frac{1}{6}$

# Primal and dual feasible bases

# Introduction: Primal and dual feasible bases

- A basis can satisfy two different conditions: non-negative values of the basic variables, or non-positive reduced costs
- We call these bases **primal feasible** and **dual feasible**: a basis that is both is optimal
- We see that a dual feasible basis gives a **feasible solution of the dual problem**, and then an upper bound on the optimal value
- This explains the whole idea of the dual simplex algorithm: it is the mirror image of the simplex algorithm

# Simplex algorithm and dual simplex algorithm

- The simplex algorithm starts from a basis associated with a **basic feasible solution**, that is, a basis with $\bar{\boldsymbol{b}} \geq 0$, and it performs pivoting operations until all the reduced costs become non-positive
- Throughout the execution, the basic solution is always feasible for the primal problem and the objective value never decreases
- The **dual simplex algorithm** does the opposite: it starts from a basis whose reduced costs are all non-positive, and it performs pivoting operations until the basic solution becomes feasible

# Primal and dual feasible bases

**Definition 1: primal feasible and dual feasible bases**

*Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard maximization form and the associated dictionary, the basis is called:*

1. ***primal feasible*** *if* $\bar{\boldsymbol{b}} \geq 0$;
2. ***dual feasible*** *if* $\bar{\boldsymbol{c}} \leq 0$.

# Primal and dual feasible: an optimal basis

- If a basis is both primal feasible and dual feasible, i.e.,

$$\bar{\boldsymbol{b}} \geq 0 \qquad \text{and} \qquad \bar{\boldsymbol{c}} \leq 0,$$

  then the associated basic feasible solution is an **optimal solution**

- Conversely, if the LP problem has an optimal solution, then there exists a basis which is both primal feasible and dual feasible

- A degenerate optimal basic solution may also be associated with a basis which is not dual feasible, i.e., with some $\bar{c}_j > 0$

# The dual solution associated with a basis (1/2)

- The name "dual feasible" is justified as follows. Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard maximization form, we associate with it the row vector of $m$ components

$$\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1},$$

which is exactly the vector appearing in the definition of the reduced costs and of the objective value of the dictionary:

$$\bar{c}_j = c_j - \underbrace{\boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=\boldsymbol{\pi}}\, \boldsymbol{A}_j, \qquad \forall j \in \mathcal{N},$$

$$\bar{z} = \underbrace{\boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=\boldsymbol{\pi}}\, \boldsymbol{b} = \boldsymbol{\pi}\, \boldsymbol{b}.$$

# The dual solution associated with a basis (2/2)

- The $m$ components of $\boldsymbol{\pi}$ are the values of the $m$ original dual variables, i.e., of the variables $\pi_{n+1}, \pi_{n+2}, \ldots, \pi_{n+m}$ of the dual problem in standard maximization form
- We now show that $\boldsymbol{\pi}$ is a feasible solution of the dual problem if and only if the basis is dual feasible
- We look first at the reduced costs of the basic variables, then at those of the non-basic variables

# The reduced costs of the basic variables are zero

- By definition of $\pi$ we have

$$\pi \, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}} \, \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{B}}}_{=\boldsymbol{I}} = \boldsymbol{c}'_{\mathcal{B}},$$

- that is, $\pi \, \boldsymbol{A}_i = c_i$ for every $i \in \mathcal{B}$,
- and then $\bar{c}_i = 0$ for every $i \in \mathcal{B}$

# The reduced costs of the non-basic variables are the dual slacks

- For every $j \in \mathcal{N}$ we have

$$\bar{c}_j \leq 0 \qquad \Longleftrightarrow \qquad c_j - \boldsymbol{\pi}\, \boldsymbol{A}_j \leq 0 \qquad \Longleftrightarrow \qquad \boldsymbol{\pi}\, \boldsymbol{A}_j \geq c_j$$

- Putting together the two previous points, the condition $\bar{\boldsymbol{c}} \leq 0$ is equivalent to

$$\boldsymbol{\pi}\, \boldsymbol{A}_j \geq c_j, \qquad \forall j \in \{1, 2, \ldots, n+m\},$$

which is precisely the set of constraints of the dual problem written for all the columns of the LP problem in standard maximization form

# The non-negativity of $\boldsymbol{\pi}$ is contained in these inequalities

- The last $m$ columns of the LP problem in standard maximization form are the columns of the slack variables, i.e.,

$$\boldsymbol{A}_{n+i} = \boldsymbol{e}_i \quad \text{with} \quad c_{n+i} = 0, \qquad \forall i \in \{1, 2, \ldots, m\}$$

- Accordingly, the inequality written for the column $n + i$ reads

$$\boldsymbol{\pi}\ \boldsymbol{e}_i \geq 0 \qquad \Longleftrightarrow \qquad \pi_i \geq 0$$

# A dual feasible basis gives an upper bound

- A basis $\boldsymbol{A}_{\mathcal{B}}$ is **dual feasible** if and only if the associated vector

$$\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1}$$

  is a **feasible solution of the dual problem**

- In this case, by the weak duality theorem, the value

$$\bar{z} = \boldsymbol{\pi} \, \boldsymbol{b}$$

  is an **upper bound** on the optimal value of the LP problem

# The mirror image of the simplex algorithm

- In the simplex algorithm every basis visited is primal feasible and then $\bar{z}$ is the value of a feasible solution of the LP problem, i.e., a lower bound on the optimal value: the algorithm increases it until it becomes also an upper bound
- In the dual simplex algorithm every basis visited is dual feasible and then $\bar{z}$ is an upper bound on the optimal value: the algorithm decreases it until it becomes also a lower bound

# The two algorithms side by side

Accordingly, the two algorithms can be described as follows:

| | **simplex algorithm** | **dual simplex algorithm** |
|---|---|---|
| starts from a basis which is | primal feasible | dual feasible |
| maintains at every iteration | primal feasibility | dual feasibility |
| terminates when the basis is also | dual feasible | primal feasible |
| or when it detects that the problem is | unbounded | infeasible |
| the value $\bar{z}$ is | a lower bound | an upper bound |
| and along the execution it is | non-decreasing | non-increasing |

Let us compute the dual solutions
associated with three bases of the running example:

the three bases visited by the dual simplex algorithm.

# Example (1/3)

- For the running example in standard maximization form, we consider the partition $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ of the slack variables. We have:

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \boldsymbol{A}_{\mathcal{B}}^{-1} \qquad \text{and} \qquad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

$$\Longrightarrow \qquad \boldsymbol{\pi} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

- which is a feasible solution of the dual problem, and $\bar{z} = \boldsymbol{\pi}\, \boldsymbol{b} = 0$ is an upper bound on the optimal value
- Consistently, the reduced costs $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} = \begin{pmatrix} -1 & -3 \end{pmatrix}$ are all non-positive

# Example (2/3)

- After the first dual pivoting operation we obtain $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$, with

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$\text{and} \qquad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \end{pmatrix},$$

- and then

$$\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

$$\bar{z} = \boldsymbol{\pi} \, \boldsymbol{b} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix} = -6.$$

# Example (3/3)

- After the second dual pivoting operation we obtain $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$, with

$$\boldsymbol{\pi} = \begin{pmatrix} \frac{7}{6} & \frac{1}{6} & 0 \end{pmatrix} \qquad \text{and} \qquad \bar{z} = \boldsymbol{\pi}\, \boldsymbol{b} = -8.$$

- The three upper bounds

$$0 \longrightarrow -6 \longrightarrow -8$$

are non-increasing, and the last one is attained by a primal feasible solution: it is then the optimal value

# Summary: Primal and dual feasible bases

**Two kinds of bases**:

- Primal feasible: $\bar{\boldsymbol{b}} \geq 0$; dual feasible: $\bar{\boldsymbol{c}} \leq 0$
- Both at the same time: the basis is optimal

**The dual solution of a basis**:

- $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \boldsymbol{A}_{\mathcal{B}}^{-1}$ is dual feasible if and only if the basis is dual feasible
- Then $\bar{z} = \boldsymbol{\pi}\, \boldsymbol{b}$ is an upper bound on the optimal value

**Dual simplex algorithm**:

- Keeps dual feasibility, looks for primal feasibility, and decreases the upper bound $\bar{z}$

# Finding an initial dual basic feasible solution

# Introduction: Finding an initial dual basic feasible solution

- The dual simplex algorithm requires in input a **dual feasible basis**, i.e., a basis whose dictionary satisfies $\bar{\boldsymbol{c}} \leq 0$
- This is the mirror image of the requirement of the simplex algorithm, which needs in input a primal feasible basis and, when none is readily available, resorts to the Phase 0
- We describe first the situations in which a dual feasible basis is available **for free**, i.e., without any computation, and then what can be done in the general case

# First case: a non-positive objective-function vector

- For an LP problem in canonical maximization form with $\boldsymbol{c} \leq 0$, the partition of the slack variables is dual feasible
- Indeed, with

$$\mathcal{B} = \{n+1, n+2, \ldots, n+m\} \quad \text{and} \quad \mathcal{N} = \{1, 2, \ldots, n\}$$

we have $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and $\boldsymbol{c}_{\mathcal{B}} = 0$, and then

$$\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} = 0$$

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \underbrace{\boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=0} \, \boldsymbol{A}_{\mathcal{N}} = \boldsymbol{c}' \leq 0$$

# Dual feasible but not primal feasible

- For an LP problem in canonical maximization form, if

$$\boldsymbol{c} \leq 0 \qquad \text{and} \qquad b_i < 0 \text{ for some } i \in \{1, 2, \ldots, m\},$$

  then the partition $\mathcal{B} = \{n+1, \ldots, n+m\}$ and $\mathcal{N} = \{1, \ldots, n\}$ of the slack variables is **dual feasible but not primal feasible**

- The dual simplex algorithm can be applied directly, while the simplex algorithm would require the Phase 0

# Second case: re-optimization after a change of $\boldsymbol{b}$ (1/2)

- Assume that an optimal basis $\boldsymbol{A}_{\mathcal{B}}$ of an LP problem is known and that the right-hand-side vector $\boldsymbol{b}$ is replaced by a new vector $\hat{\boldsymbol{b}}$
- The reduced costs

$$\bar{c}_j = c_j - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_j, \qquad \forall j \in \mathcal{N},$$

do not depend on $\boldsymbol{b}$, and then they remain all non-positive: the basis **remains dual feasible**

# Second case: re-optimization after a change of $\boldsymbol{b}$ (2/2)

- On the contrary, the values of the basic variables

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\ \hat{\boldsymbol{b}}$$

  do change and may become negative: the basis may lose primal feasibility

- The dual simplex algorithm can then be started from the old optimal basis and typically restores optimality in very few iterations

# Third case: adding a constraint to an optimal basis (1/4)

- Assume that an optimal basis $\boldsymbol{A}_{\mathcal{B}}$ of an LP problem is known, with optimal solution $\tilde{\boldsymbol{x}}$, and that a new constraint

$$\boldsymbol{a}' \boldsymbol{x} \leq \beta$$

  is added to the problem, together with its slack variable $x_{n+m+1}$

- We consider the new partition

$$\hat{\mathcal{B}} = \mathcal{B} \cup \{n+m+1\} \quad \text{and} \quad \hat{\mathcal{N}} = \mathcal{N}$$

# Third case: adding a constraint to an optimal basis (2/4)

a. The new basis matrix and its inverse are

$$\boldsymbol{A}_{\hat{\mathcal{B}}} = \begin{pmatrix} \boldsymbol{A}_{\mathcal{B}} & 0 \\ \boldsymbol{a}'_{\mathcal{B}} & 1 \end{pmatrix}$$

$$\text{and} \qquad \boldsymbol{A}_{\hat{\mathcal{B}}}^{-1} = \begin{pmatrix} \boldsymbol{A}_{\mathcal{B}}^{-1} & 0 \\ -\boldsymbol{a}'_{\mathcal{B}}\,\boldsymbol{A}_{\mathcal{B}}^{-1} & 1 \end{pmatrix},$$

where $\boldsymbol{a}'_{\mathcal{B}}$ collects the components of $\boldsymbol{a}'$ associated with the basic variables

In particular, $\boldsymbol{A}_{\hat{\mathcal{B}}}$ is invertible and then it is a basis

# Third case: adding a constraint to an optimal basis (3/4)

b. Since the new slack variable has objective-function coefficient equal to zero, we have $\boldsymbol{c}'_{\hat{\mathcal{B}}} = \begin{pmatrix} \boldsymbol{c}'_{\mathcal{B}} & 0 \end{pmatrix}$ and then

$$\hat{\boldsymbol{\pi}} = \boldsymbol{c}'_{\hat{\mathcal{B}}} \, \boldsymbol{A}_{\hat{\mathcal{B}}}^{-1} = \begin{pmatrix} \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} & 0 \end{pmatrix} = \begin{pmatrix} \boldsymbol{\pi} & 0 \end{pmatrix},$$

that is, the dual variable of the new constraint is set to zero and all the other dual variables are unchanged

Accordingly, for every $j \in \hat{\mathcal{N}}$,

$$\hat{c}_j = c_j - \begin{pmatrix} \boldsymbol{\pi} & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{A}_j \\ a_j \end{pmatrix} = c_j - \boldsymbol{\pi} \, \boldsymbol{A}_j = \bar{c}_j \leq 0 :$$

the reduced costs are unchanged and the new basis is **dual feasible**

# Third case: adding a constraint to an optimal basis (4/4)

c. The value of the new basic variable is

$$\bar{b}_{n+m+1} = \beta - \boldsymbol{a}' \tilde{\boldsymbol{x}},$$

which is negative if and only if the old optimal solution $\tilde{\boldsymbol{x}}$ violates the new constraint

In this case the dual simplex algorithm is started from $\hat{\mathcal{B}}$

# Fourth case: branch-and-bound and cutting planes

- The third case is exactly what happens when an integer linear programming problem is solved: at every node of the branch-and-bound tree a bound such as

$$x_j \leq \lfloor \tilde{x}_j \rfloor \qquad \text{or} \qquad x_j \geq \lceil \tilde{x}_j \rceil$$

  is added to the LP relaxation solved at the parent node, and the same happens every time a cutting plane is added

- The optimal basis of the parent node is dual feasible for the child node, and the dual simplex algorithm re-optimizes it with a handful of pivoting operations instead of solving the new LP problem from scratch

- This is one of the main reasons why the dual simplex algorithm is among the most used algorithms in modern LP solvers, and the one typically used to re-optimize inside branch-and-bound

Let us see the second and the third case
on the running example:

first we change the right-hand-side vector,
then we add a new constraint.

# Example: re-optimization after a change of $\boldsymbol{b}$ (1/6)

- We consider the running example and its optimal basis $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$, with the optimal dictionary:

$$\begin{cases} x_1 = 2 + \frac{2}{3} x_3 - \frac{1}{3} x_4 \\ x_2 = 2 + \frac{1}{6} x_3 + \frac{1}{6} x_4 \\ x_5 = 3 - x_3 + x_4 \\ z = -8 - \frac{7}{6} x_3 - \frac{1}{6} x_4 \end{cases}$$

# Example: re-optimization after a change of $\boldsymbol{b}$ (2/6)

- Assume now that the right-hand side of the first constraint changes from $-6$ to $-12$, i.e., that

$$\boldsymbol{b} = \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix} \qquad \longrightarrow \qquad \hat{\boldsymbol{b}} = \begin{pmatrix} -12 \\ -6 \\ 3 \end{pmatrix}$$

- The reduced costs do not depend on the right-hand-side vector, and then $\bar{\boldsymbol{c}}' = \begin{pmatrix} -\frac{7}{6} & -\frac{1}{6} \end{pmatrix}$ is unchanged and the basis remains **dual feasible**

# Example: re-optimization after a change of $\boldsymbol{b}$ (3/6)

- The values of the basic variables, instead, become

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\hat{\boldsymbol{b}} = \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} & 0 \\ -\frac{1}{6} & -\frac{1}{6} & 0 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} -12 \\ -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ -3 \end{pmatrix}$$

# Example: re-optimization after a change of $\boldsymbol{b}$ (4/6)

- The dictionary becomes:

$$\begin{cases} x_1 = 6 + \frac{2}{3} x_3 - \frac{1}{3} x_4 \\ x_2 = 3 + \frac{1}{6} x_3 + \frac{1}{6} x_4 \\ x_5 = -3 - x_3 + x_4 \\ z = -15 - \frac{7}{6} x_3 - \frac{1}{6} x_4 \end{cases}$$

- The basis is no longer primal feasible, since $\bar{b}_5 = -3 < 0$

# Example: re-optimization after a change of $\boldsymbol{b}$ (5/6)

- We then perform a dual pivoting operation with pivot row $x_5$
- The values of its row are $\bar{a}_{53} = 1 \geq 0$ and $\bar{a}_{54} = -1 < 0$, and then $x_4$ is the only candidate and it is the pivot column, with

$$s_m = \frac{\bar{c}_4}{\bar{a}_{54}} = \frac{-\frac{1}{6}}{-1} = \frac{1}{6}$$

- Updating the dictionary, we obtain $\mathcal{B} = \{1, \ \ 2, \ \ 4\}$ and $\mathcal{N} = \{3, \ \ 5\}$

# Example: re-optimization after a change of $\boldsymbol{b}$ (6/6)

- The updated dictionary is:

$$\begin{cases} x_1 = 5 + \frac{1}{3}x_3 - \frac{1}{3}x_5 \\ x_2 = \frac{7}{2} + \frac{1}{3}x_3 + \frac{1}{6}x_5 \\ x_4 = 3 + x_3 + x_5 \\ z = -\frac{31}{2} - \frac{4}{3}x_3 - \frac{1}{6}x_5 \end{cases}$$

- Now $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$: after a single dual pivoting operation the new optimal solution is

$$\tilde{\boldsymbol{x}}(LP) = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix} \qquad \text{and} \qquad z(LP) = -\frac{31}{2}$$

# Example: adding a constraint to an optimal basis (1/6)

- We consider again the running example with its optimal basis $\mathcal{B} = \{1, \ 2, \ 5\}$ and its optimal solution $\tilde{x}_1 = 2$ and $\tilde{x}_2 = 2$
- We add the new constraint

$$x_1 + x_2 \leq 3,$$

  together with its slack variable $x_6$
- The new basis is $\hat{\mathcal{B}} = \{1, \ 2, \ 5, \ 6\}$ and $\hat{\mathcal{N}} = \{3, \ 4\}$

# Example: adding a constraint to an optimal basis (2/6)

- The reduced costs are unchanged, and then the new basis is dual feasible
- The value of the new basic variable is

$$\bar{b}_6 = 3 - (\tilde{x}_1 + \tilde{x}_2) = 3 - 4 = -1 < 0,$$

since the old optimal solution violates the new constraint

- The equation of $x_6$ in the new dictionary is obtained by substituting the equations of $x_1$ and $x_2$:

$$\begin{aligned} x_6 &= 3 - x_1 - x_2 \\ &= 3 - \left(2 + \frac{2}{3}x_3 - \frac{1}{3}x_4\right) - \left(2 + \frac{1}{6}x_3 + \frac{1}{6}x_4\right) \\ &= -1 - \frac{5}{6}x_3 + \frac{1}{6}x_4 \end{aligned}$$

# Example: adding a constraint to an optimal basis (3/6)

- The new dictionary is:

$$\begin{cases} x_1 = 2 + \frac{2}{3} x_3 - \frac{1}{3} x_4 \\ x_2 = 2 + \frac{1}{6} x_3 + \frac{1}{6} x_4 \\ x_5 = 3 - x_3 + x_4 \\ x_6 = -1 - \frac{5}{6} x_3 + \frac{1}{6} x_4 \\ z = -8 - \frac{7}{6} x_3 - \frac{1}{6} x_4 \end{cases}$$

# Example: adding a constraint to an optimal basis (4/6)

- We perform a dual pivoting operation with pivot row $x_6$
- The values of its row are $\bar{a}_{63} = \frac{5}{6} \geq 0$ and $\bar{a}_{64} = -\frac{1}{6} < 0$, and then $x_4$ is the pivot column, with

$$s_m = \frac{\bar{c}_4}{\bar{a}_{64}} = \frac{-\frac{1}{6}}{-\frac{1}{6}} = 1$$

- Updating the dictionary, we obtain $\mathcal{B} = \{1, \ 2, \ 4, \ 5\}$ and $\mathcal{N} = \{3, \ 6\}$

# Example: adding a constraint to an optimal basis (5/6)

- The updated dictionary is:

$$\begin{cases} x_1 = 0 - x_3 - 2x_6 \\ x_2 = 3 + x_3 + x_6 \\ x_4 = 6 + 5x_3 + 6x_6 \\ x_5 = 9 + 4x_3 + 6x_6 \\ z = -9 - 2x_3 - x_6 \end{cases}$$

# Example: adding a constraint to an optimal basis (6/6)

- Again a single dual pivoting operation restores optimality, and the new optimal solution is

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} \qquad \text{and} \qquad z(\,LP\,) = -9$$

- The new optimal value $-9$ is not larger than the old one $-8$, as it must be, since adding a constraint to a maximization problem can only reduce the optimal value

# The general case

- When neither $\boldsymbol{b} \geq 0$ nor $\boldsymbol{c} \leq 0$, the partition of the slack variables is neither primal feasible nor dual feasible, and no basis is available for free
- Besides the Phase 0, the dual simplex algorithm offers an alternative: we change the objective function, without changing the feasible region

# Finding a primal feasible basis with the dual simplex algorithm (1/2)

a. Replace the objective-function vector $\boldsymbol{c}$ by any vector $\hat{\boldsymbol{c}} \leq 0$, for instance $\hat{\boldsymbol{c}} = 0$ or $\hat{c}_j = \min\{c_j, 0\}$. The feasible region is unchanged.

b. The partition of the slack variables is dual feasible for the modified problem. Run the dual simplex algorithm on it, with the smallest-index rule (Bland's rule) described later in this chapter.

c. If it terminates with the certificate of infeasibility, the modified problem is infeasible and then the original problem is infeasible too, since they have the same feasible region. `STOP`.

# Finding a primal feasible basis with the dual simplex algorithm (2/2)

d. Otherwise it terminates with a basis $\mathcal{B}$ which is primal feasible for the modified problem, and then
also for the original problem.

e. Restore the original vector $\boldsymbol{c}$, recompute the reduced costs $\bar{\boldsymbol{c}}$ and the value $\bar{z}$ of the basis $\mathcal{B}$, and run the simplex algorithm starting from it.

# An alternative to the Phase 0

- This is a genuine alternative to the Phase 0: it plays exactly the same role, i.e., it provides an initial primal feasible basis or a proof that none exists
- It does so without introducing the auxiliary variable $x_0$ and without solving an auxiliary problem
- With $\hat{\boldsymbol{c}} = 0$ all the reduced costs are equal to zero at every iteration, all the ratios of the ratio test are equal to zero and the pivot column is always chosen by the tie-breaking part of Bland's rule alone, i.e., as the candidate with the smallest index
- Since Bland's rule prevents cycling, the procedure terminates

# Example: finding a primal feasible basis (1/8)

- We consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrrrrrrr} \max & 3 & x_1 & + & & x_2 & & \\ \text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\ & & x_1 & + & 2 & x_2 & \leq & 8 \\ & 2 & x_1 & + & & x_2 & \leq & 10 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

- Here $\boldsymbol{c} \not\leq 0$ and $\boldsymbol{b} \not\geq 0$: the partition $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ of the slack variables is neither primal feasible nor dual feasible

# Example: finding a primal feasible basis (2/8)

- **Step a.** We replace the objective-function vector by $\hat{\boldsymbol{c}} = 0$. The dictionary of the modified problem is:

$$\begin{cases} x_3 = -2 + x_1 + x_2 \\ x_4 = 8 - x_1 - 2\,x_2 \\ x_5 = 10 - 2\,x_1 - x_2 \\ \hat{z} = 0 + 0\,x_1 + 0\,x_2 \end{cases}$$

- All the reduced costs are non-positive, and then the basis is dual feasible

# Example: finding a primal feasible basis (3/8)

- **Step b.** The pivot row is $x_3$, since $\bar{b}_3 = -2 < 0$
- The values of its row are $\bar{a}_{31} = -1 < 0$ and $\bar{a}_{32} = -1 < 0$, and then both non-basic variables are candidates:

$$s_m = \min \left\{ \underbrace{\frac{0}{-1}}_{=0}, \underbrace{\frac{0}{-1}}_{=0} \right\} = 0,$$

  and the two ratios are equal

- By Bland's rule we choose the smallest index, i.e., $x_1$ is the pivot column

# Example: finding a primal feasible basis (4/8)

- Updating the dictionary, with $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$, we obtain:

$$\begin{cases} x_1 = 2 - x_2 + x_3 \\ x_4 = 6 - x_2 - x_3 \\ x_5 = 6 + x_2 - 2x_3 \\ \hat{z} = 0 + 0x_2 + 0x_3 \end{cases}$$

- **Step d.** Now $\bar{\boldsymbol{b}} = \begin{pmatrix} 2 & 6 & 6 \end{pmatrix}' \geq 0$: the basis $\mathcal{B} = \{1, \ 4, \ 5\}$ is **primal feasible**, both for the modified problem and for the original one

# Example: finding a primal feasible basis (5/8)

- **Step e.** We restore the original objective function and we recompute the objective-function row by substituting the equation of $x_1$:

$$z = 3\,x_1 + x_2 = 3\,(2 - x_2 + x_3) + x_2 = 6 - 2\,x_2 + 3\,x_3,$$

which gives the dictionary:

$$\begin{cases} x_1 &= 2 - x_2 + x_3 \\ x_4 &= 6 - x_2 - x_3 \\ x_5 &= 6 + x_2 - 2\,x_3 \\ z &= 6 - 2\,x_2 + 3\,x_3 \end{cases}$$

# Example: finding a primal feasible basis (6/8)

- The reduced cost $\bar{c}_3 = 3 > 0$, and then we perform a (primal) pivoting operation with pivot column $x_3$
- We find the largest $t$ such that $\bar{\boldsymbol{b}} - t\,\bar{\boldsymbol{A}}_3 \geq 0$:

$$\begin{cases} \underbrace{2 - t \cdot (-1) \geq 0}_{x_1 \geq 0} \\ \underbrace{6 - t \cdot 1 \geq 0}_{x_4 \geq 0} \\ \underbrace{6 - t \cdot 2 \geq 0}_{x_5 \geq 0} \end{cases} \qquad \begin{cases} t \geq -2 \\ t \leq 6 \\ t \leq 3 \end{cases} \qquad \text{then} \quad t_m = 3$$

- The pivot row is $x_5$

# Example: finding a primal feasible basis (7/8)

- Updating the dictionary, with $\mathcal{B} = \{1, \ 3, \ 4\}$ and $\mathcal{N} = \{2, \ 5\}$, we obtain:

$$\begin{cases} x_1 = 5 - \frac{1}{2} x_2 - \frac{1}{2} x_5 \\ x_3 = 3 + \frac{1}{2} x_2 - \frac{1}{2} x_5 \\ x_4 = 3 - \frac{3}{2} x_2 + \frac{1}{2} x_5 \\ z = 15 - \frac{1}{2} x_2 - \frac{3}{2} x_5 \end{cases}$$

# Example: finding a primal feasible basis (8/8)

- All the reduced costs are non-positive: the basis is optimal and

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 5 \\ 0 \end{pmatrix} \qquad \text{and} \qquad z(\,LP\,) = 15$$

- The dual simplex algorithm found the primal feasible basis $\{1, 4, 5\}$ with one degenerate dual pivoting operation; the simplex algorithm then reached the optimum with one pivoting operation

# Summary: initial dual feasible basis (1/2)

**A dual feasible basis for free**:

- $\boldsymbol{c} \leq 0$: the partition of the slack variables is dual feasible
- After a change of $\boldsymbol{b}$: the reduced costs do not change, the old optimal basis stays dual feasible
- After adding a constraint: the old optimal basis plus the new slack variable is dual feasible, with $\bar{b}_{n+m+1} = \beta - \boldsymbol{a}' \tilde{\boldsymbol{x}}$

**Re-optimization**:

- Branch-and-bound and cutting planes add constraints: the dual simplex algorithm re-optimizes with a few dual pivoting operations

# Summary: initial dual feasible basis (2/2)

**The general case**:

- Replace $\boldsymbol{c}$ by $\hat{\boldsymbol{c}} \leq 0$: the feasible region does not change
- Run the dual simplex algorithm with Bland's rule: it finds a primal feasible basis or proves that the problem is infeasible
- Restore $\boldsymbol{c}$ and continue with the simplex algorithm

**An alternative to the Phase** 0:

- Same role, no auxiliary variable $x_0$ and no auxiliary problem

# Dual pivoting operations

# Introduction: Dual pivoting operations

- A dual pivoting operation is the single step of the dual simplex algorithm: it moves from a dual feasible basis to a new dual feasible basis
- We see how to choose the pivot row and the pivot column, and why the new basis stays dual feasible
- We see the rule that breaks the ties (Bland's rule), the certificate of infeasibility and the update of the dictionary
- Everything is shown on the running example

# When is a dual pivoting operation needed?

- Given a dual feasible basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard maximization form, if in the associated dictionary there exists $r \in \mathcal{B}$ such that $\bar{b}_r < 0$ (a negative basic value), it is necessary to perform a dual pivoting operation
- The dual simplex algorithm works on the primal dictionary, exactly the same object used by the simplex algorithm
- What changes is the **order** of the two choices

# Dual pivoting operation

**Definition 2: dual pivoting operation**

*A **dual pivoting operation** consists in determining a **new dual feasible basis** with the goal of restoring the non-negativity of one basic variable, by **swapping one basic variable and one non-basic variable**.*

- *The basic variable that becomes non-basic is called the **pivot row**, and it is chosen first;*
- *The non-basic variable that becomes basic is called the **pivot column**, and it is chosen second.*

- The order of the two choices is the opposite of the one of the simplex algorithm

# Determining the pivot row (1/2)

- **The pivot row is a basic variable with a negative value**, i.e., a basic variable

$$x_r \quad \text{with} \quad r \in \mathcal{B} \quad \text{such that} \quad \bar{b}_r < 0$$

- The variable $x_r$ leaves the basis and becomes non-basic, and then it takes the value 0: this is precisely what restores its non-negativity

# Determining the pivot row (2/2)

- If several basic variables have a negative value, any one of them preserves dual feasibility. In these notes we always choose the one with the **smallest index**:

$$r = \min \left\{ \; i \in \mathcal{B} : \bar{b}_i < 0 \; \right\}$$

- LP solvers often choose instead the most negative value $\bar{b}_r$, or more refined criteria

- The smallest-index choice is part of Bland's rule (see later in this section), which guarantees that the algorithm terminates

# Determining the pivot column (1/5)

- The pivot column is chosen so that dual feasibility is preserved, that is, so that all the reduced costs remain non-positive after the dual pivoting operation. We now derive the rule

- The equation of the pivot row in the dictionary and the equation of the objective function are

$$x_r = \bar{b}_r - \sum_{j \in \mathcal{N}} \bar{a}_{rj}\, x_j \qquad \text{and} \qquad z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j\, x_j$$

- The first one can be rewritten as the identity

$$0 = \bar{b}_r - \sum_{j \in \mathcal{N}} \bar{a}_{rj}\, x_j - x_r,$$

which holds for every point satisfying the equality constraints

# Determining the pivot column (2/5)

- Accordingly, for every value $s \in \mathbb{R}$, we can add $s$ times this identity to the equation of the objective function without changing it:

$$z = \underbrace{\bar{z} + s\ \bar{b}_r}_{\text{new } \bar{z}} + \sum_{j \in \mathcal{N}} \underbrace{\left(\bar{c}_j - s\ \bar{a}_{rj}\right)}_{\text{new } \bar{c}_j} x_j + \underbrace{(-s)}_{\text{new } \bar{c}_r} x_r \qquad (1)$$

- These $|\mathcal{N}| + 1$ coefficients are the reduced costs of the new dictionary, in which $x_r$ has become non-basic

- **The value of $s$ is increased as much as possible while at the same time maintaining non-positive all the reduced costs**. In this way a new dual feasible basis is determined

# Determining the pivot column (3/5)

- Imposing the non-positivity of all the coefficients of (1), we obtain:

$$\begin{cases} -s \leq 0 \\ \bar{c}_j - s\,\bar{a}_{rj} \leq 0, \qquad \forall j \in \mathcal{N} \end{cases} \implies$$

$$\begin{cases} s \geq 0 \\ s \leq \dfrac{\bar{c}_j}{\bar{a}_{rj}}, \qquad \forall j \in \mathcal{N} \text{ with } \bar{a}_{rj} < 0 \end{cases}$$

- The inequalities with $\bar{a}_{rj} > 0$ are satisfied for every $s \geq 0$, since $\bar{c}_j \leq 0$
- Those with $\bar{a}_{rj} = 0$ reduce to $\bar{c}_j \leq 0$, which holds by hypothesis
- The direction of the inequality is reversed when dividing by $\bar{a}_{rj} < 0$

# Determining the pivot column (4/5)

- Accordingly, the value of $s$ becomes the maximum value, called $s_m$, such that:

$$s_m = \min \left\{ \frac{\bar{c}_j}{\bar{a}_{rj}} : j \in \mathcal{N},\ \bar{a}_{rj} < 0 \right\} \geq 0$$

- With this value of $s$, the new reduced cost of the variable $x_p$ attaining the minimum becomes

$$\bar{c}_p - s_m\, \bar{a}_{rp} = \bar{c}_p - \frac{\bar{c}_p}{\bar{a}_{rp}}\, \bar{a}_{rp} = 0,$$

and a variable whose reduced cost is zero can enter the basis: $x_p$ is the pivot column

# Determining the pivot column (5/5)

- **The pivot column is a non-basic variable whose reduced cost becomes zero by setting the dual step to $s_m$**, i.e., a non-basic variable

$$x_p \quad \text{with} \quad p \in \mathcal{N} \quad \text{such that}$$

$$\frac{\bar{c}_p}{\bar{a}_{rp}} = s_m = \min \left\{ \frac{\bar{c}_j}{\bar{a}_{rj}} : j \in \mathcal{N},\ \bar{a}_{rj} < 0 \right\},$$

where $\bar{a}_{rp}$ is the value of the row $\bar{\boldsymbol{a}}_r$ associated with the pivot column $x_p$

# The new reduced costs and the new objective value

- Given $s_m$ and the row $\bar{\boldsymbol{a}}_r$, **the new reduced costs** (which determine the new vector $\bar{\boldsymbol{c}}$) and **the new objective value** are obtained as follows:

$$\begin{aligned} \bar{c}_j &\leftarrow \bar{c}_j - s_m\, \bar{a}_{rj}, \qquad \forall j \in \mathcal{N} \setminus \{p\}, \\ \bar{c}_r &\leftarrow -s_m, \\ \bar{c}_p &\leftarrow 0, \\ \bar{z} &\leftarrow \bar{z} + s_m\, \bar{b}_r \end{aligned} \tag{2}$$

# The objective value never increases

- Since $s_m \geq 0$ and $\bar{b}_r < 0$, we have

$$\bar{z} + s_m \, \bar{b}_r \;\leq\; \bar{z}$$

- **The objective value never increases**, which is the mirror image of what happens in the simplex algorithm, where the objective value never decreases

- By the previous section, $\bar{z}$ is at every iteration an upper bound on the optimal value, and the dual simplex algorithm decreases it

# The new partition of the indices

- Given a pivot row $x_r$ with $r \in \mathcal{B}$ and a pivot column $x_p$ with $p \in \mathcal{N}$, **the new partition of the indices of the variables** is updated as follows:

$$\mathcal{B} \leftarrow \left(\mathcal{B} \setminus \{r\}\right) \cup \{p\} \quad \text{and} \quad \mathcal{N} \leftarrow \left(\mathcal{N} \setminus \{p\}\right) \cup \{r\}$$

- By construction $\boldsymbol{A}_{\mathcal{B}}$ remains a basis, since $\bar{a}_{rp} \neq 0$

# Ties and degenerate dual pivoting operations

- The pivot row and the pivot column are not always unique: several basic variables may have a negative value, and several non-basic variables may attain the minimum $s_m$
- If $s_m = 0$ (a **degenerate** dual pivoting operation) the value $\bar{z}$ does not change and, with an arbitrary choice, the algorithm could cycle
- In these notes we always break the ties with the smallest-index rule

# Bland's rule for the dual simplex algorithm

a. The pivot row is the basic variable with negative value having the **smallest index**:

$$r = \min \left\{ \; i \in \mathcal{B} : \bar{b}_i < 0 \; \right\}$$

b. The pivot column is the candidate attaining $s_m$ having the **smallest index**:

$$p = \min \left\{ \; j \in \mathcal{N} : \bar{a}_{rj} < 0, \; \frac{\bar{c}_j}{\bar{a}_{rj}} = s_m \; \right\}$$

- By the correspondence $\pi_j \leftrightarrow x_j$, this is exactly Bland's rule of the simplex algorithm applied to the dual problem, and then the dual simplex algorithm with this rule always terminates

# When the ratio test has no candidate

- If $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$, then the ratio test has no candidate
- The value of $s$ can be increased without bound while maintaining non-positive all the reduced costs
- In this case the LP problem is infeasible

# Certificate of infeasibility

- Given a dual feasible basis $\boldsymbol{A}_{\mathcal{B}}$, if there exists a basic variable $x_r$ with $r \in \mathcal{B}$ such that

$$\bar{b}_r < 0 \qquad \text{and} \qquad \bar{a}_{rj} \geq 0, \quad \forall j \in \mathcal{N},$$

  then the LP problem in standard maximization form is **infeasible**

- We give two proofs: one on the primal problem and one on the dual problem

# First proof (on the primal)

- The equation of the pivot row in the dictionary is

$$x_r = \bar{b}_r - \sum_{j \in \mathcal{N}} \bar{a}_{rj}\, x_j,$$

  and every feasible solution satisfies $x_j \geq 0$ for every $j \in \mathcal{N}$
- Since $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$, every term of the sum is non-negative and we obtain

$$x_r \leq \bar{b}_r < 0,$$

  which contradicts the non-negativity of $x_r$
- Accordingly, no feasible solution exists

□

# Second proof (on the dual)

- For every value $s \geq 0$, the coefficients of (1) are all non-positive, and then they are the reduced costs of a dual feasible solution of value

$$\bar{z} + s\ \bar{b}_r$$

- Since $\bar{b}_r < 0$, this value tends to $-\infty$ as $s$ grows: the dual problem is **unbounded**

- By the weak duality theorem, the primal problem is then **infeasible**

□

This is the exact mirror image of the certificate of unboundedness of the simplex algorithm.

# Update of the dictionary

Once the pivot row $x_r$ with $r \in \mathcal{B}$
and the pivot column $x_p$ with $p \in \mathcal{N}$ are determined,

the new dictionary is obtained with
**exactly the same row operations as in the simplex algorithm**.

# The equation of the pivot column

- Let $x_p$ be the entering variable and $x_r$ the leaving variable. From the equation of the pivot row, we have:

$$x_r = \bar{b}_r - \bar{a}_{rp} x_p - \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{a}_{rj} x_j$$

- Solving for $x_p$ gives the equation of the pivot column in the new dictionary:

$$x_p = \frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r \tag{3}$$

# The other rows of the new dictionary

- For each remaining basic variable $i \in \mathcal{B} \setminus \{r\}$, substituting with (3) gives their equations in the new dictionary
- Substituting in the equation of the objective function gives the new objective-function row, whose coefficients are exactly those of (2)
- The new value of the pivot column is

$$\bar{b}_p = \frac{\bar{b}_r}{\bar{a}_{rp}} \geq 0,$$

since $\bar{b}_r < 0$ and $\bar{a}_{rp} < 0$: the variable entering the basis takes a non-negative value

- The values of the other basic variables, instead, may well be negative, and further dual pivoting operations are then needed

Let us perform one dual pivoting operation
on the running example:

first the choice of the pivot row and of the pivot column,
then the update of the dictionary.

# Example: dual pivoting operations (1/3)

- For the running example, with $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$, we have the dictionary:

$$\begin{cases} x_3 &= -6 &+ & x_1 &+ & 2\,x_2 \\ x_4 &= -6 &- & x_1 &+ & 4\,x_2 \\ x_5 &= 3 &- & 2\,x_1 &+ & 2\,x_2 \\ z &= 0 &- & x_1 &- & 3\,x_2 \end{cases}$$

- All the reduced costs are non-positive, and then the basis is dual feasible
- Two basic variables have a negative value, $\bar{b}_3 = -6 < 0$ and $\bar{b}_4 = -6 < 0$, and by the smallest-index rule we choose $x_3$ as the pivot row

# Example: dual pivoting operations (2/3)

- We then find the largest $s$ such that all the reduced costs of (1) remain non-positive:

$$\begin{cases} \underbrace{-s \leq 0}_{\bar{c}_3 \leq 0} \\ \underbrace{-1 - s \cdot (-1) \leq 0}_{\bar{c}_1 \leq 0} \\ \underbrace{-3 - s \cdot (-2) \leq 0}_{\bar{c}_2 \leq 0} \end{cases} \qquad \begin{cases} s \geq 0 \\ s \leq 1 \\ s \leq \frac{3}{2} \end{cases}$$

$$\text{then} \quad s_m = \min \left\{ \underbrace{\frac{-1}{-1}}_{=1}, \underbrace{\frac{-3}{-2}}_{=\frac{3}{2}} \right\} = 1$$

# Example: dual pivoting operations (3/3)

- The dual step becomes $s_m = 1$, the reduced cost of $x_1$ becomes zero and $x_1$ is the pivot column
- The other reduced costs and the objective value become $\bar{c}_j - s_m \bar{a}_{3j}$ and $\bar{z} + s_m \bar{b}_3$, then we have:

$$\bar{c}_2 = -3 - 1 \cdot (-2) = -1, \quad \bar{c}_3 = -1, \quad \bar{z} = 0 + 1 \cdot (-6) = -6,$$

and accordingly

$$\begin{pmatrix} \bar{c}_2 \\ \bar{c}_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- The new partition of the indices of the variables is $\mathcal{B} = \{1, 4, 5\}$ and $\mathcal{N} = \{2, 3\}$

# Example: update of the dictionary (1/2)

- We perform the dual pivoting operation of the previous example, with $x_3$ as the pivot row and $x_1$ as the pivot column:

$$x_3 = -6 + x_1 + 2\,x_2 \quad \text{then} \quad x_1 = 6 - 2\,x_2 + x_3$$

- Substituting in the other equations of the dictionary, we get:

$$\begin{aligned} x_4 &= -6 - x_1 + 4\,x_2 = -6 - (6 - 2\,x_2 + x_3) + 4\,x_2 \\ &= -12 + 6\,x_2 - x_3 \end{aligned}$$

$$\begin{aligned} x_5 &= 3 - 2\,x_1 + 2\,x_2 = 3 - 2\,(6 - 2\,x_2 + x_3) + 2\,x_2 \\ &= -9 + 6\,x_2 - 2\,x_3 \end{aligned}$$

$$\begin{aligned} z &= -\,x_1 - 3\,x_2 = -\,(6 - 2\,x_2 + x_3) - 3\,x_2 \\ &= -6 - x_2 - x_3 \end{aligned}$$

# Example: update of the dictionary (2/2)

- The new dictionary, with $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$, is:

$$\begin{cases} x_1 = 6 - 2x_2 + x_3 \\ x_4 = -12 + 6x_2 - x_3 \\ x_5 = -9 + 6x_2 - 2x_3 \\ z = -6 - x_2 - x_3 \end{cases}$$

- The reduced costs and the objective value coincide with those computed in the previous example, as expected
- The value of the entering variable is $\bar{b}_1 = \frac{\bar{b}_3}{\bar{a}_{31}} = \frac{-6}{-1} = 6 \geq 0$

# Summary: Dual pivoting operations (1/2)

**Pivot row** (chosen first):

- A basic variable with $\bar{b}_r < 0$; with Bland's rule, the one with the smallest index

**Pivot column** (chosen second):

- The ratio test $s_m = \min\{\bar{c}_j / \bar{a}_{rj} : j \in \mathcal{N},\ \bar{a}_{rj} < 0\}$ keeps all the reduced costs non-positive
- Ties: the candidate with the smallest index

**Objective value**:

- $\bar{z} \leftarrow \bar{z} + s_m \bar{b}_r \leq \bar{z}$: the upper bound never increases

# Summary: Dual pivoting operations (2/2)

**Certificate of infeasibility**:

- $\bar{b}_r < 0$ and $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$: the LP problem is infeasible (and the dual problem is unbounded)

**Update of the dictionary**:

- The same row operations as in the simplex algorithm
- The entering variable takes the value $\bar{b}_r / \bar{a}_{rp} \geq 0$; the other basic values may stay negative

# Pseudo-code of the dual simplex algorithm

# Introduction: Pseudo-code of the dual simplex algorithm

- We put together the pieces: the feasibility check, the choice of the pivot row and of the pivot column, the certificate of infeasibility and the update of the dictionary
- The result is the pseudo-code of the dual simplex algorithm, with Bland's rule
- We execute it on two examples: the running example, which has an optimal solution, and an infeasible LP problem
- Along the execution we also follow the dual solutions, which are vertices of the dual polyhedron

# The dual simplex algorithm (1/4)

**Input**

An LP problem in standard maximization form and a **dual feasible** basis, i.e., a basis whose dictionary satisfies $\bar{\boldsymbol{c}} \leq 0$.

**Output**

An optimal solution of the LP problem, if one exists; otherwise, a certificate that the LP problem is infeasible.

Step 0: **Initialization**

Consider the given dual feasible basis and the associated dictionary. Go to Step 1.

# The dual simplex algorithm (2/4)

Step 1: **Feasibility check**

With respect to the current dictionary, if

$$\bar{b}_i \geq 0, \qquad \forall i \in \mathcal{B},$$

then the basis is primal feasible and dual feasible, the basic feasible solution $x_i = \bar{b}_i$, $\forall i \in \mathcal{B}$, and $x_j = 0$, $\forall j \in \mathcal{N}$, is an optimal solution and $\bar{z}$ is the optimal value. In this case `STOP`.

# The dual simplex algorithm (3/4)

Step 2: **Dual pivoting operations**

The **pivot row** is the basic variable $x_r$ with negative value having the smallest index:

$$r = \min \left\{ \; i \in \mathcal{B} : \bar{b}_i < 0 \; \right\}.$$

a. If

$$\bar{a}_{rj} \geq 0, \qquad \forall j \in \mathcal{N},$$

then the LP problem is **infeasible**. `STOP`.

# The dual simplex algorithm (4/4)

Step 2: **Dual pivoting operations** (continued)

b. Determine the **pivot column**, i.e., a non-basic variable $x_p$ with $p \in \mathcal{N}$ such that

$$\frac{\bar{c}_p}{\bar{a}_{rp}} = s_m = \min \left\{ \frac{\bar{c}_j}{\bar{a}_{rj}} : j \in \mathcal{N},\ \bar{a}_{rj} < 0 \right\};$$

in case of ties, choose the candidate with the smallest index $p$.

c. Update the dictionary with the same row operations as in the simplex algorithm. In particular, $\bar{z} \leftarrow \bar{z} + s_m \bar{b}_r$.

d. Update the basis:

$$\mathcal{B} \leftarrow \big(\mathcal{B} \setminus \{r\}\big) \cup \{p\} \quad \text{and} \quad \mathcal{N} \leftarrow \big(\mathcal{N} \setminus \{p\}\big) \cup \{r\}.$$

Go to Step 1.

Let us execute the dual simplex algorithm
on the running example,
starting from the partition of the slack variables.

# Example: execution of the dual simplex algorithm (1/9)

- Consider the running example in standard maximization form. Its objective-function vector satisfies $\boldsymbol{c} \leq 0$, and then the partition $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ of the slack variables is **dual feasible**
- The associated dictionary is:

$$\begin{cases} x_3 = -6 + x_1 + 2x_2 \\ x_4 = -6 - x_1 + 4x_2 \\ x_5 = 3 - 2x_1 + 2x_2 \\ z = 0 - x_1 - 3x_2 \end{cases}$$

- All the reduced costs are non-positive, but the basic solution is <u>not</u> feasible, since $\bar{b}_3 = -6 < 0$ and $\bar{b}_4 = -6 < 0$

# Example: execution of the dual simplex algorithm (2/9)

- **First dual pivoting operation.** The pivot row is $x_3$, the basic variable with negative value having the smallest index

- The values of its row are $\bar{a}_{31} = -1 < 0$ and $\bar{a}_{32} = -2 < 0$, and then both non-basic variables are candidates:

$$\min \left\{ \underbrace{\frac{-1}{-1}}_{=1}, \underbrace{\frac{-3}{-2}}_{=\frac{3}{2}} \right\} = s_m = 1$$

  and the pivot column is $x_1$

- The new objective value is $\bar{z} + s_m \bar{b}_3 = 0 + 1 \cdot (-6) = -6$

# Example: execution of the dual simplex algorithm (3/9)

- The new dictionary, with $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$, is:

$$\begin{cases} x_1 &= 6 - 2x_2 + x_3 \\ x_4 &= -12 + 6x_2 - x_3 \\ x_5 &= -9 + 6x_2 - 2x_3 \\ z &= -6 - x_2 - x_3 \end{cases}$$

- The reduced costs are still all non-positive, as expected, but the basic solution is still not feasible

# Example: execution of the dual simplex algorithm (4/9)

- **Second dual pivoting operation.** The pivot row is $x_4$, the basic variable with negative value having the smallest index ($\bar{b}_4 = -12$, $\bar{b}_5 = -9$)

- The only negative value of its row is $\bar{a}_{42} = -6$, and then the pivot column is $x_2$, with dual step

$$s_m = \frac{\bar{c}_2}{\bar{a}_{42}} = \frac{-1}{-6} = \frac{1}{6}$$

- The new objective value is

$$\bar{z} + s_m \, \bar{b}_4 = -6 + \frac{1}{6} \cdot (-12) = -8$$

# Example: execution of the dual simplex algorithm (5/9)

- The new dictionary, with $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$, is:

$$\begin{cases} x_1 = 2 + \frac{2}{3} x_3 - \frac{1}{3} x_4 \\ x_2 = 2 + \frac{1}{6} x_3 + \frac{1}{6} x_4 \\ x_5 = 3 - x_3 + x_4 \\ z = -8 - \frac{7}{6} x_3 - \frac{1}{6} x_4 \end{cases}$$

- Now $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$: the basis is primal feasible and dual feasible, and then it is optimal

# Example: execution of the dual simplex algorithm (6/9)

- An optimal solution of the running example is

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

  and the optimal value is $z(\,LP\,) = -8$, in accordance with the solution found at the beginning of the chapter

- The three bases visited by the algorithm correspond to three feasible solutions of the **dual** problem, given by $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\,\boldsymbol{A}_{\mathcal{B}}^{-1}$:

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \longrightarrow \begin{pmatrix} \frac{7}{6} \\ \frac{1}{6} \\ 0 \end{pmatrix}$$

# Example: execution of the dual simplex algorithm (7/9)

- The dual objective values $\boldsymbol{b}'\boldsymbol{\pi}$ of these solutions are

$$0 \longrightarrow -6 \longrightarrow -8,$$

  non-increasing, as it must be for a minimization problem

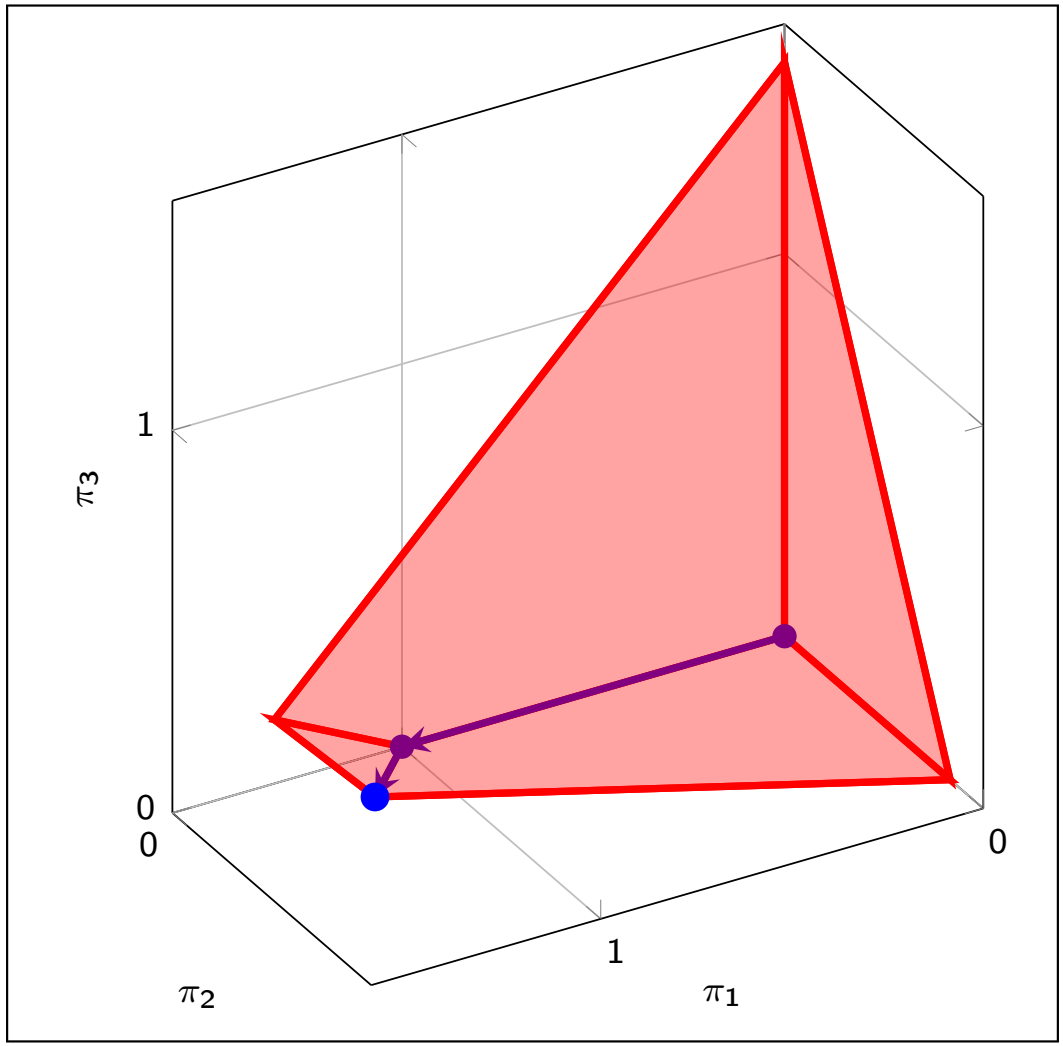
- Here they even decrease strictly, but a degenerate dual pivoting operation with $s_m = 0$ leaves the value unchanged

- They are vertices of the dual polyhedron, and the dual simplex algorithm moves from one of them to an adjacent one: it is the simplex algorithm executed on the dual problem

# Example: execution of the dual simplex algorithm (8/9)

- In the figure on the next slide the axes are the three original dual variables, called $\pi_1, \pi_2, \pi_3$ in the dual problem of the running example
- Recalling that in the standard maximization form their indices are shifted by $n = 2$, they correspond to $\pi_3, \pi_4, \pi_5$
- In violet the first two dual vertices visited by the algorithm, in blue the optimal one
- The corresponding primal basic solutions are instead not feasible until the last iteration

# Example: execution of the dual simplex algorithm (9/9)



Let us now see what happens
when the LP problem has no feasible solution:

the dual simplex algorithm stops
with the certificate of infeasibility.

# Example: certificate of infeasibility (1/9)

- Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrrrrrrr} \max & -2 & x_1 & - & & x_2 & & \\ \text{subject to} & - & x_1 & - & 2 & x_2 & \leq & -6 \\ & -2 & x_1 & - & & x_2 & \leq & -6 \\ & & x_1 & + & & x_2 & \leq & 2 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

- Since $\boldsymbol{c} \leq 0$, the partition $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ is dual feasible

# Example: certificate of infeasibility (2/9)

- The initial dictionary is:

$$\begin{cases} x_3 = -6 + x_1 + 2x_2 \\ x_4 = -6 + 2x_1 + x_2 \\ x_5 = 2 - x_1 - x_2 \\ z = 0 - 2x_1 - x_2 \end{cases}$$

# Example: certificate of infeasibility (3/9)

- **First dual pivoting operation.** By the smallest-index rule the pivot row is $x_3$, whose value is $\bar{b}_3 = -6 < 0$
- The values of its row are $\bar{a}_{31} = -1 < 0$ and $\bar{a}_{32} = -2 < 0$, and then both non-basic variables are candidates:

$$\min \left\{ \underbrace{\frac{-2}{-1}}_{=2}, \underbrace{\frac{-1}{-2}}_{=\frac{1}{2}} \right\} = \frac{1}{2}$$

and the pivot column is $x_2$

# Example: certificate of infeasibility (4/9)

- The new dictionary, with $\mathcal{B} = \{2, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 3\}$, is:

$$\begin{cases} x_2 = 3 - \frac{1}{2} x_1 + \frac{1}{2} x_3 \\ x_4 = -3 + \frac{3}{2} x_1 + \frac{1}{2} x_3 \\ x_5 = -1 - \frac{1}{2} x_1 - \frac{1}{2} x_3 \\ z = -3 - \frac{3}{2} x_1 - \frac{1}{2} x_3 \end{cases}$$

# Example: certificate of infeasibility (5/9)

- **Second dual pivoting operation.** Two basic variables have a negative value, $x_4$ and $x_5$: by the smallest-index rule the pivot row is $x_4$

- The values of its row are $\bar{a}_{41} = -\frac{3}{2} < 0$ and $\bar{a}_{43} = -\frac{1}{2} < 0$, and then both non-basic variables are candidates:

$$\min \left\{ \underbrace{\frac{-\frac{3}{2}}{-\frac{3}{2}}}_{=1}, \underbrace{\frac{-\frac{1}{2}}{-\frac{1}{2}}}_{=1} \right\} = 1$$

- The two ratios are equal and, by the smallest-index rule, the pivot column is $x_1$

# Example: certificate of infeasibility (6/9)

- We have:

$$x_4 = -3 + \frac{3}{2}x_1 + \frac{1}{2}x_3 \quad \text{then} \quad x_1 = 2 - \frac{1}{3}x_3 + \frac{2}{3}x_4,$$

and, substituting in the other equations of the dictionary:

$$x_2 = 3 - \frac{1}{2}\left(2 - \frac{1}{3}x_3 + \frac{2}{3}x_4\right) + \frac{1}{2}x_3 = 2 + \frac{2}{3}x_3 - \frac{1}{3}x_4$$

$$x_5 = -1 - \frac{1}{2}\left(2 - \frac{1}{3}x_3 + \frac{2}{3}x_4\right) - \frac{1}{2}x_3 = -2 - \frac{1}{3}x_3 - \frac{1}{3}x_4$$

$$z = -3 - \frac{3}{2}\left(2 - \frac{1}{3}x_3 + \frac{2}{3}x_4\right) - \frac{1}{2}x_3 = -6 + 0\,x_3 - x_4$$

# Example: certificate of infeasibility (7/9)

- The new dictionary, with $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$, is:

$$\begin{cases} x_1 = 2 - \frac{1}{3} x_3 + \frac{2}{3} x_4 \\ x_2 = 2 + \frac{2}{3} x_3 - \frac{1}{3} x_4 \\ x_5 = -2 - \frac{1}{3} x_3 - \frac{1}{3} x_4 \\ z = -6 + 0 x_3 - x_4 \end{cases}$$

- Because of the tie in the ratio test, the reduced cost $\bar{c}_3$ becomes 0 too: it is still non-positive and the basis is still dual feasible, with $\bar{z} = -3 + 1 \cdot (-3) = -6$

# Example: certificate of infeasibility (8/9)

- **Third dual pivoting operation.** The only basic variable with a negative value is $x_5$, whose row has values

$$\bar{a}_{53} = \frac{1}{3} \geq 0 \qquad \text{and} \qquad \bar{a}_{54} = \frac{1}{3} \geq 0,$$

that is, all non-negative

- By the certificate of infeasibility, the LP problem is **infeasible**

- Indeed, the equation of $x_5$ in this dictionary reads

$$x_5 = -2 - \frac{1}{3} x_3 - \frac{1}{3} x_4,$$

and every feasible solution satisfies $x_3 \geq 0$ and $x_4 \geq 0$, which gives $x_5 \leq -2 < 0$, in contradiction with the non-negativity of $x_5$

# Example: certificate of infeasibility (9/9)

- Accordingly, no feasible solution exists
- The certificate was already available in the previous dictionary, where the row of $x_5$ was

$$x_5 = -1 - \frac{1}{2}x_1 - \frac{1}{2}x_3$$

with all non-negative values $\bar{a}_{51}, \bar{a}_{53}$: the smallest-index rule selected $x_4$ first

- A certificate of infeasibility can be read in any row with $\bar{b}_r < 0$ and $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$, even if it is not the pivot row

# Summary: Pseudo-code of the dual simplex algorithm

**The algorithm**:

- Input: a dual feasible basis; output: an optimal solution or a certificate of infeasibility
- Step 1: if $\bar{\boldsymbol{b}} \geq 0$, the basis is optimal
- Step 2: pivot row, certificate check, pivot column, update of the dictionary and of the basis

**The two examples**:

- Running example: two dual pivoting operations, upper bounds $0 \to -6 \to -8$, optimal value $-8$; the dual solutions are vertices of the dual polyhedron
- Infeasible LP problem: a row with $\bar{b}_r < 0$ and all $\bar{a}_{rj} \geq 0$ proves that no feasible solution exists

# Solving the dual problem with the simplex algorithm

# Introduction: Solving the dual problem with the simplex algorithm

- In this section we execute the **simplex algorithm on the dual problem** of the running example and, in parallel, we report at every iteration the corresponding dictionary of the primal problem
- The two executions are two different ways of describing the same sequence of bases
- The dual simplex algorithm applied to the primal problem is exactly the simplex algorithm applied to the dual problem

# Example: initialization (1/4)

- We shift the indices of the original dual variables by $n = 2$ and we add $n = 2$ surplus variables in front. We then obtain the following standard maximization form:

$$
\begin{array}{lrrrrrrrrr}
\text{max} & & & & & 6\,\pi_3 & + & 6\,\pi_4 & - & 3\,\pi_5 \\
\text{subject to} & -\,\pi_1 & & - & & \pi_3 & + & \pi_4 & + & 2\,\pi_5 = -1 \\
& & - & \pi_2 & - & 2\,\pi_3 & - & 4\,\pi_4 & - & 2\,\pi_5 = -3 \\
& \pi_1, & & \pi_2, & & \pi_3, & & \pi_4, & & \pi_5 \geq 0
\end{array}
$$

# Example: initialization (2/4)

- The dual initial dictionary is:

$$\begin{cases} \pi_1 = & 1 & - & & \pi_3 & + & & \pi_4 & + & 2 & \pi_5 \\ \pi_2 = & 3 & - & 2 & \pi_3 & - & 4 & \pi_4 & - & 2 & \pi_5 \\ -w = & 0 & + & 6 & \pi_3 & + & 6 & \pi_4 & - & 3 & \pi_5 \end{cases}$$

- where $\mathcal{B} = \{1, 2\}$ and $\mathcal{N} = \{3, 4, 5\}$. The basic dual feasible solution is:

$$\bar{\pi}_1 = 1, \quad \bar{\pi}_2 = 3, \qquad \bar{\pi}_3 = 0, \quad \bar{\pi}_4 = 0, \quad \bar{\pi}_5 = 0$$

with $-w = 0$, i.e., $w = 0$

# Example: initialization (3/4)

- After having added the slack variables to the primal problem, we obtain the following standard form:

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & & & & & & & & \\
\text{subject to} & - & x_1 & - & 2 & x_2 & + & x_3 & & & & & = & -6 \\
 & & x_1 & - & 4 & x_2 & & & + & x_4 & & & = & -6 \\
 & 2 & x_1 & - & 2 & x_2 & & & & & + & x_5 & = & 3 \\
 & & x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

# Example: initialization (4/4)

- The primal initial dictionary is:

$$\begin{cases} x_3 = -6 + x_1 + 2x_2 \\ x_4 = -6 - x_1 + 4x_2 \\ x_5 = 3 - 2x_1 + 2x_2 \\ z = 0 - x_1 - 3x_2 \end{cases}$$

- where $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$. The basic primal non-feasible solution is:

$$\bar{x}_3 = -6, \quad \bar{x}_4 = -6, \quad \bar{x}_5 = 3, \qquad \bar{x}_1 = 0, \quad \bar{x}_2 = 0$$

of value 0

# Example: iteration 1 (1/7)

- If we increase the value of the variable $\pi_3$, we can get a better dual solution. We have:

$$\begin{cases} \pi_1 = 1 - \pi_3 \\ \\ \pi_2 = 3 - 2\,\pi_3 \end{cases} \qquad \begin{cases} \underbrace{1 - \pi_3 \geq 0}_{\pi_1 \geq 0} \\ \\ \underbrace{3 - 2\,\pi_3 \geq 0}_{\pi_2 \geq 0} \end{cases} \qquad \begin{cases} \pi_3 \leq 1 \\ \\ \pi_3 \leq \frac{3}{2} \end{cases}$$

- By increasing the value of $\pi_3$ to 1, the value of $\pi_1$ decreases to 0

# Example: iteration 1 (2/7)

- Then we have that:

  $\pi_3$ becomes dual basic and, accordingly, $x_3$ becomes primal non-basic;

  $\pi_1$ becomes dual non-basic and, accordingly, $x_1$ becomes primal basic.

- The dual pivoting operation of the dual simplex algorithm chose exactly $x_3$ as the pivot row and $x_1$ as the pivot column

# Example: iteration 1 (3/7)

- **For the dual LP problem**, we have:

$$\pi_1 = 1 - \pi_3 + \pi_4 + 2\,\pi_5 \quad \text{then} \quad \pi_3 = 1 - \pi_1 + \pi_4 + 2\,\pi_5.$$

- Substituting in the other equations of the dual dictionary, we get:

$$\begin{aligned} \pi_2 &= 3 - 2\,(1 - \pi_1 + \pi_4 + 2\,\pi_5) - 4\,\pi_4 - 2\,\pi_5 \\ &= 1 + 2\,\pi_1 - 6\,\pi_4 - 6\,\pi_5 \\ -w &= 0 + 6\,\,(1 - \pi_1 + \pi_4 + 2\,\pi_5) + 6\,\pi_4 - 3\,\pi_5 \\ &= 6 - 6\,\pi_1 + 12\,\pi_4 + 9\,\pi_5 \end{aligned}$$

# Example: iteration 1 (4/7)

- Accordingly, we now have the following dual dictionary:

$$\begin{cases} \pi_2 = 1 + 2\pi_1 - 6\pi_4 - 6\pi_5 \\ \pi_3 = 1 - \pi_1 + \pi_4 + 2\pi_5 \\ -w = 6 - 6\pi_1 + 12\pi_4 + 9\pi_5 \end{cases}$$

- where $\mathcal{B} = \{2, 3\}$ and $\mathcal{N} = \{1, 4, 5\}$. The basic dual feasible solution is:

$$\bar{\pi}_2 = 1, \quad \bar{\pi}_3 = 1, \qquad \bar{\pi}_1 = 0, \quad \bar{\pi}_4 = 0, \quad \bar{\pi}_5 = 0,$$

with $-w = 6$, i.e., $w = -6$

# Example: iteration 1 (5/7)

- **For the primal LP problem**, we have:

$$x_3 = -6 + x_1 + 2\,x_2 \quad \text{then} \quad x_1 = 6 - 2x_2 + x_3.$$

- Substituting in the other equations of the primal dictionary, we get:

$$x_4 = -6 - (6 - 2x_2 + x_3) + 4\,x_2 = -12 + 6\,x_2 - x_3$$

$$x_5 = 3 - 2\,(6 - 2x_2 + x_3) + 2\,x_2 = -9 + 6\,x_2 - 2\,x_3$$

$$z = 0 - (6 - 2x_2 + x_3) - 3\,x_2 = -6 - x_2 - x_3$$

# Example: iteration 1 (6/7)

- Accordingly, we now have the following primal dictionary:

$$\begin{cases} x_1 = & 6 & - & 2\,x_2 & + & x_3 \\ x_4 = & -12 & + & 6\,x_2 & - & x_3 \\ x_5 = & -9 & + & 6\,x_2 & - & 2\,x_3 \\ z = & -6 & - & x_2 & - & x_3 \end{cases}$$

- where $\mathcal{B} = \{1, 4, 5\}$ and $\mathcal{N} = \{2, 3\}$

# Example: iteration 1 (7/7)

- The basic primal non-feasible solution is:

$$\bar{x}_1 = 6, \quad \bar{x}_4 = -12, \quad \bar{x}_5 = -9, \qquad \bar{x}_2 = 0, \quad \bar{x}_3 = 0$$

  of value $-6$

- The primal dictionary is exactly the one obtained by the first dual pivoting operation of the dual simplex algorithm

# Example: iteration 2 (1/7)

- If we increase the value of the variable $\pi_4$, we can get a better dual solution. We have:

$$\begin{cases} \pi_2 = 1 - 6\,\pi_4 \\ \\ \pi_3 = 1 + \pi_4 \end{cases} \qquad \begin{cases} \underbrace{1 - 6\,\pi_4 \geq 0}_{\pi_2 \geq 0} \\ \\ \underbrace{1 + \pi_4 \geq 0}_{\pi_3 \geq 0} \end{cases} \qquad \begin{cases} \pi_4 \leq \frac{1}{6} \\ \\ \pi_4 \geq -1 \end{cases}$$

- By increasing the value of $\pi_4$ to $\frac{1}{6}$, the value of $\pi_2$ decreases to 0

# Example: iteration 2 (2/7)

- Then we have that:

  $\pi_4$ becomes dual basic and, accordingly, $x_4$ becomes primal non-basic;

  $\pi_2$ becomes dual non-basic and, accordingly, $x_2$ becomes primal basic.

- The dual pivoting operation of the dual simplex algorithm chose exactly $x_4$ as the pivot row and $x_2$ as the pivot column

# Example: iteration 2 (3/7)

- **For the dual LP problem**, we have:

$$\pi_2 = 1 + 2\,\pi_1 - 6\,\pi_4 - 6\,\pi_5$$

$$\text{then} \quad \pi_4 = \frac{1}{6} + \frac{1}{3}\,\pi_1 - \frac{1}{6}\,\pi_2 - \pi_5.$$

- Substituting in the other equations of the dual dictionary, we get:

$$\begin{aligned}
\pi_3 &= 1 - \pi_1 + \left(\frac{1}{6} + \frac{1}{3}\,\pi_1 - \frac{1}{6}\,\pi_2 - \pi_5\right) + 2\,\pi_5 \\
&= \frac{7}{6} - \frac{2}{3}\,\pi_1 - \frac{1}{6}\,\pi_2 + \pi_5 \\
-w &= 6 - 6\,\pi_1 + 12\,\left(\frac{1}{6} + \frac{1}{3}\,\pi_1 - \frac{1}{6}\,\pi_2 - \pi_5\right) + 9\,\pi_5 \\
&= 8 - 2\,\pi_1 - 2\,\pi_2 - 3\,\pi_5
\end{aligned}$$

# Example: iteration 2 (4/7)

- Accordingly, we now have the following optimal dual dictionary:

$$\begin{cases} \pi_3 = \frac{7}{6} - \frac{2}{3}\pi_1 - \frac{1}{6}\pi_2 + \pi_5 \\ \pi_4 = \frac{1}{6} + \frac{1}{3}\pi_1 - \frac{1}{6}\pi_2 - \pi_5 \\ -w = 8 - 2\pi_1 - 2\pi_2 - 3\pi_5 \end{cases}$$

- where $\mathcal{B} = \{3, 4\}$ and $\mathcal{N} = \{1, 2, 5\}$. The dual basic optimal solution is:

$$\tilde{\pi}_3 = \frac{7}{6}, \quad \tilde{\pi}_4 = \frac{1}{6}, \qquad \tilde{\pi}_1 = 0, \quad \tilde{\pi}_2 = 0, \quad \tilde{\pi}_5 = 0,$$

with $-w = 8$, i.e., $w = -8$

# Example: iteration 2 (5/7)

- **For the primal LP problem**, we have:

$$x_4 = -12 + 6\,x_2 - x_3 \quad \text{then} \quad x_2 = 2 + \frac{1}{6}\,x_3 + \frac{1}{6}\,x_4.$$

- Substituting in the other equations of the dictionary, we get:

$$x_1 = 6 - 2\left(2 + \frac{1}{6}\,x_3 + \frac{1}{6}\,x_4\right) + x_3 = 2 + \frac{2}{3}\,x_3 - \frac{1}{3}\,x_4$$

$$x_5 = -9 + 6\left(2 + \frac{1}{6}\,x_3 + \frac{1}{6}\,x_4\right) - 2\,x_3 = 3 - x_3 + x_4$$

$$z = -6 - \left(2 + \frac{1}{6}\,x_3 + \frac{1}{6}\,x_4\right) - x_3 = -8 - \frac{7}{6}\,x_3 - \frac{1}{6}\,x_4$$

# Example: iteration 2 (6/7)

- Accordingly, we now have the following optimal primal dictionary:

$$\begin{cases} x_1 = & 2 & + & \frac{2}{3} x_3 & - & \frac{1}{3} x_4 \\ x_2 = & 2 & + & \frac{1}{6} x_3 & + & \frac{1}{6} x_4 \\ x_5 = & 3 & - & x_3 & + & x_4 \\ z = & -8 & - & \frac{7}{6} x_3 & - & \frac{1}{6} x_4 \end{cases}$$

- where $\mathcal{B} = \{1, 2, 5\}$ and $\mathcal{N} = \{3, 4\}$

# Example: iteration 2 (7/7)

- The primal basic optimal solution is:

$$\tilde{x}_1 = 2, \quad \tilde{x}_2 = 2, \quad \tilde{x}_5 = 3, \qquad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0$$

  of value $-8$

- The two executions visited the same bases and stopped at the same optimal value: $w = -8$ for the dual problem and $z = -8$ for the primal problem

# Summary: Solving the dual problem with the simplex algorithm

**Two descriptions of the same execution**:

- The simplex algorithm on the dual problem: $\pi_j$ enters the dual basis exactly when $x_j$ leaves the primal basis, and conversely
- The primal dictionaries are those of the dual simplex algorithm on the running example

**Values along the execution**:

- Dual: $w = 0 \to -6 \to -8$; primal: $\bar{z} = 0 \to -6 \to -8$
- The primal basic solutions become feasible only at the last iteration

# Summary (1/2)

- A basis is **primal feasible** if $\bar{\boldsymbol{b}} \geq 0$ and **dual feasible** if $\bar{\boldsymbol{c}} \leq 0$; a basis that is both is optimal
- A dual feasible basis gives the dual feasible solution $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}} \boldsymbol{A}_{\mathcal{B}}^{-1}$, and $\bar{z} = \boldsymbol{\pi}\, \boldsymbol{b}$ is an upper bound on the optimal value
- A dual feasible basis is available for free when $\boldsymbol{c} \leq 0$, after a change of $\boldsymbol{b}$ and after adding a constraint; in the general case the dual simplex algorithm with $\hat{\boldsymbol{c}} \leq 0$ replaces the Phase 0

# Summary (2/2)

- A **dual pivoting operation** chooses first the pivot row ($\bar{b}_r < 0$) and then the pivot column (ratio test $s_m$), keeps dual feasibility and never increases $\bar{z}$
- If the row of $x_r$ has $\bar{b}_r < 0$ and all $\bar{a}_{rj} \geq 0$, the LP problem is **infeasible**
- With Bland's rule the dual simplex algorithm always terminates
- The dual simplex algorithm on the primal problem is the **simplex algorithm on the dual problem**: it moves along adjacent vertices of the dual polyhedron